Spectral Arithmetic Warp: Vacuum Engineering via the Bost-Connes Phase Transition and Euler Product Amplification

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[1] *Independent Research*

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We develop five approaches to manipulating quantum vacuum energy using the arithmetic structure of the Riemann zeta function, as realized in the Bost-Connes (BC) C*-dynamical system. (1) *Spectral action approach*: local perturbation δD of the Dirac operator changes geometry via Connes' spectral action principle, with the BC heat kernel $\mathrm{Tr}(e^{-tH}) = \zeta(t)$ providing the bridge between number theory and gravity. (2) *β-pumping*: population inversion of cavity phonons creates a transient $\beta < 0$ state (negative temperature), realizing local dark energy ($\beta = -1$) via a standard laboratory technique (Purcell-Pound protocol). (3) *Euler product amplification*: removing K primes from the mode spectrum amplifies Casimir energy by $\prod_{i=1}^{K}(p_i^3 - 1)$; 14 primes yield $10^{48}\times$. (4) *Inverse spectral design*: a graded material whose natural eigenfrequencies match the prime-sieved set, physically instantiating the Euler product as an "arithmetic drum." (5) *Dirac spectral modulation*: π-phase rotation of all Euler factors (Liouville function vacuum, $\zeta(t)/\zeta(2t)$) yields $G_{\mathrm{eff}} = -2G$ (repulsive gravity); muting $K \geq 2$ primes yields $G_{\mathrm{eff}} = 0$ (gravity nullification). A combined two-layer warp bubble (outer: repulsive, inner: gravity-free) requires $\sim 10^3$ J of positive energy—44 orders of magnitude below Alcubierre. We prove that rapid switching yields exactly zero net energy (vacuum Carnot theorem), confirming energy conservation. The decisive experimental test—comparing Casimir energy in a 2-prime-sieved cavity versus a uniform cavity (predicted ratio 182:1 vs. 1:3)—costs approximately \$2,200.

I. INTRODUCTION

The Alcubierre warp metric [?] requires negative energy density (violation of the weak energy condition), typically estimated at $\sim 10^{47}$ J of exotic matter. This paper explores whether the arithmetic structure of the quantum vacuum—specifically the Euler product decomposition of the Riemann zeta function—offers alternative mechanisms for vacuum engineering.

Our starting point is the Bost-Connes (BC) C*-dynamical system [?], whose partition function is the Riemann zeta function:

$$Z(\beta) = \mathrm{Tr}(e^{-\beta H}) = \sum_{n=1}^{\infty} n^{-\beta} = \zeta(\beta). \tag{1}$$

The BC system exhibits a phase transition at $\beta = 1$, with spontaneous breaking of the Galois symmetry $\mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q})$ for $\beta > 1$.

In previous work [?], we showed that the dark energy density $\Omega_\Lambda = 2\pi/9$ arises from the completed zeta function at negative temperature:

$$\Omega_\Lambda = \frac{d}{\mathrm{cd}}\,|\xi_{\neg 2}(-1)| = \frac{4}{3}\cdot\frac{\pi}{6} = \frac{2\pi}{9} \approx 0.698, \tag{2}$$

where $d = 4$, $\mathrm{cd} = \mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3$ (Artin-Verdier), and $\xi_{\neg 2}$ is the completed zeta with the $p = 2$ Euler factor removed.

Here we ask: *can the same arithmetic structure be exploited for local vacuum engineering?*

II. THE β-LINE OF PHYSICS

The inverse temperature β of the BC system parameterizes distinct physical regimes:

β	$\zeta(\beta)$	Physical sector
$\beta > 1$	convergent	Perturbative QFT
$\beta = 1$	pole	Phase transition
$0 < \beta < 1$		Grand unification
$\beta = 0$	$-1/2$	Topological
$\beta = -1$	$-1/12$	Dark energy
$\beta = -3$	$1/120$	3D Casimir / K-theory

The functional equation $\xi(\beta) = \xi(1-\beta)$ establishes a duality: $\beta = -1 \leftrightarrow \beta = 2$, mapping dark energy to perturbative physics.

The sign of $\zeta_{\neg 2}$ alternates at odd negative integers: $\zeta_{\neg 2}(-1) = +1/12$ (positive, repulsive), $\zeta_{\neg 2}(-3) = -7/120$ (negative), establishing a dimension-dependent sign structure.

III. APPROACH 1: SPECTRAL ACTION

Connes' spectral action principle [?] gives

$$S = \mathrm{Tr}\big(f(D^2/\Lambda^2)\big) \sim \sum_k f_k \Lambda^k a_k(D^2), \tag{3}$$

where a_k are the Seeley-DeWitt coefficients encoding geometry.

For $D^2 = H_{\rm BC}$ with heat kernel (1): $S(\Lambda) = \zeta(1/\Lambda^2)$. The pole at $\Lambda = 1$ ($\leftrightarrow \beta = 1$) is the BC phase transition, where the spectral action diverges.

A local perturbation $D \to D + V(x)$ modifies the spectral action density:

$$s(x) = \zeta(1/\Lambda^2)\, e^{-V(x)/\Lambda^2}, \tag{4}$$

producing expansion ($V < 0$) or contraction ($V > 0$)—an Alcubierre-like profile in spectral language.

Proposition 1 (Finite energy, divergent response)**.** *Near the BC phase transition $\beta \to 1^+$: the energy cost $E(\beta) = -\zeta'(\beta)/\zeta(\beta)$ remains finite, while the geometric response $\zeta(\beta) \sim 1/(\beta - 1)$ diverges.*

However, the specific heat $C(\beta)$ also diverges as $1/(\beta - 1)^2$, limiting the leverage ratio χ/C to $O(1)$. The product $\delta g \times E_{\rm wall}$ scales identically to general relativity, confirming that the spectral approach does not circumvent energy-geometry proportionality by itself.

IV. APPROACH 2: β-PUMPING

The $\beta = -1$ state (dark energy) can be created locally via *population inversion*—a standard technique in NMR and quantum optics (Purcell-Pound, 1951).

Protocol. (1) Prepare a BAW crystal in DD thermal equilibrium at temperature T. (2) Rapidly switch one boundary from D to N (switching time $\tau < 1/(2\omega_0)$). (3) The phonon populations are now inverted relative to the new mode spectrum: $\beta_{\rm eff} < 0$. (4) During the relaxation time T_1, the cavity exhibits negative effective pressure.

The energy density of the population-inverted state is $\rho_{\rm dyn} \sim N_{\rm modes}\, k_B T/V$, which exceeds the static Casimir energy $\rho_{\rm Cas} \sim \hbar c/d^4$ by a factor $\sim \bar{n} \times N_{\rm modes} \sim 10^{20}$.

The effect is *transient* (duration $\sim T_1$) but *repeatable* (pulsed operation at $\sim 1/(2T_1)$). This constitutes a propellantless acoustic thruster concept, though with small forces ($\sim 10^{-13}$ N per crystal).

V. APPROACH 3: EULER PRODUCT AMPLIFICATION

The Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ decomposes the vacuum into independent prime channels. Removing primes $p_1, \ldots, p_K$ gives

$$\zeta_{\neg\{p_i\}}(s) = \prod_{i=1}^{K} (1 - p_i^{-s})\, \zeta(s). \tag{5}$$

At $s = -3$ (3D Casimir), each removed prime amplifies the spectral zeta by $|1 - p^3| = p^3 - 1$:

K	Primes	Amplification
1	$\{2\}$	$7\times$
2	$\{2, 3\}$	$182\times$
4	$\{2, 3, 5, 7\}$	7.7×10^6
14	$\{2, \ldots, 43\}$	1.9×10^{48}

For even K, the Casimir energy is negative (exotic matter). At $K = 14$, the amplification exceeds the Alcubierre requirement.

A. Safety analysis

Previous work identified that removing primes from the *global* vacuum changes L-function values linked to coupling constants (θ_W, α_s). However, *local* mode filtering in a cavity—analogous to the Casimir effect between plates, which modifies the mode spectrum without changing α—may not affect global coupling constants.

This distinction is *not rigorously established*. The BC framework suggests L-function values change under Euler factor removal, while standard QFT UV/IR decoupling suggests they do not.

Conjecture 2 (Regulator conjecture)**.** *For a physical cavity whose spectral zeta function equals $\zeta_{\neg\{p_i\}}(s)$, the Casimir energy equals the zeta-regularized value $-(\hbar c/d^3) \times \zeta_{\neg\{p_i\}}(-3) \times$ (geometric factor).*

This conjecture is experimentally testable (Section IX).

B. Vacuum Carnot theorem

Theorem 3 (Energy conservation under rapid switching)**.** *A cycle of boundary-condition switching* DD $\to$ DN$_{\neg 2,3}$ $\to$ DD *with intermediate mechanical work yields exactly zero net energy:*

$$\oint \delta E = 0 \tag{6}$$

for any path in (L, BC) parameter space.

Proof. The total energy is $E(L, \mathrm{BC}) = -A_{\rm BC}/L^3$ where $A_{\rm BC}$ depends only on the boundary condition. In a cycle: $\Delta E_{\rm switch} + W_{\rm mech} = 0$ at each step, and the sum telescopes to zero. □

This rules out energy extraction from vacuum switching and confirms the theoretical self-consistency of the framework.

VI. APPROACH 4: INVERSE SPECTRAL DESIGN

Rather than suppressing modes with bandgap filters (*subtractive*), we propose designing a graded material

whose *natural* eigenfrequencies match the prime-sieved set (*constructive*).

The inverse Sturm-Liouville problem: given target eigenvalues $\lambda_k = m_k^2$ where $\{m_k\}$ is the prime-sieved sequence, find a potential $V(x)$ on $[0, \pi]$ such that

$$-y'' + V(x)y = \lambda y, \quad y(0) = y(\pi) = 0 \tag{7}$$

has spectrum $\{\lambda_k\}$.

By the Gel'fand-Levitan theorem, $V(x)$ exists and is unique (given appropriate normalizing constants). The potential encodes the Eratosthenes sieve in its shape.

Material realization. $V(x)$ maps to position-dependent stiffness $c(x)^2$ in an acoustic resonator. This can be fabricated as a compositionally graded piezoelectric thin film (e.g., $\mathrm{Al}_x\mathrm{Ga}_{1-x}\mathrm{N}$ with programmed $x(z)$) via molecular beam epitaxy or atomic layer deposition.

The *arithmetic drum*—a single graded crystal—replaces the 14-period phononic sieve, with no bandgap leakage and exact eigenvalues by construction.

VII. APPROACH 5: DIRAC SPECTRAL MODULATION

Rather than removing Euler factors (muting) or designing specific eigenfrequencies (inverse spectral), we consider *phase-rotating* the Euler factors: replacing $(1 - p^{-t})^{-1}$ with $(1 - e^{i\phi}p^{-t})^{-1}$.

A. Gravity nullification ($G_{\text{eff}} = 0$)

The Seeley-DeWitt coefficient a_2 (which determines G) is proportional to the heat kernel derivative $K'(0)$. When K primes are muted, $\zeta_{\neg\{p_i\}}(t)$ has a K-fold zero at $t = 0$:

$$\zeta_{\neg\{p_i\}}(t) = \prod_{i=1}^{K}(1 - p_i^{-t})\,\zeta(t) \sim t^K \quad (t \to 0). \tag{8}$$

For $d = 4$, the heat kernel expansion $K(t) = a_0 t^{-2} + a_2 t^{-1} + a_4 + \cdots$ requires $a_0 = a_2 = 0$ when $K \geq 2$.

Proposition 4 (Gravity nullification)**.** *If the spectral zeta of a local region equals $\zeta_{\neg\{p_1,p_2\}}(t)$ for any two primes p_1, p_2, then $a_0 = a_2 = 0$ in that region, implying $G_{\text{eff}} = 0$ and no cosmological constant.*

B. Anti-gravity ($G_{\text{eff}} < 0$) via Liouville rotation

Setting $\phi = \pi$ for *all* primes replaces each $(1 - p^{-t})^{-1}$ with $(1 + p^{-t})^{-1}$. Using the identity

$$\prod_p (1 + p^{-t})^{-1} = \frac{\zeta(2t)}{\zeta(t)}, \tag{9}$$

which follows from $\zeta(t)/\zeta(2t) = \prod_p(1 + p^{-t})$, the all-rotated heat kernel is $K_\pi(t) = \zeta(t)/\zeta(2t)$.

At $t = 0$: $K_\pi(0) = \zeta(0)/\zeta(0) = 1$ (no zero). The derivative:

$$K'_\pi(0) = \frac{d}{dt}\left[\frac{\zeta(t)}{\zeta(2t)}\right]_{t=0} = -\frac{\zeta'(0)}{\zeta(0)} = \log(2\pi). \tag{10}$$

Compare the normal case $K'(0) = \zeta'(0) = -\frac{1}{2}\log(2\pi)$. The ratio

$$\frac{G_{\text{eff}}}{G} = \frac{K'_\pi(0)}{K'(0)} = \frac{\log(2\pi)}{-\frac{1}{2}\log(2\pi)} = -2. \tag{11}$$

Proposition 5 (Anti-gravity)**.** *If all prime Euler factors are π-rotated (Liouville function vacuum), the effective gravitational coupling reverses: $G_{\text{eff}} = -2G$. This is repulsive gravity.*

Implementation. Each transmon qubit (tuned to $\omega_p = \log p/\tau$) applies a Z-gate (π-phase shift) rather than absorbing the mode. The Z-gate is a standard single-qubit operation in circuit QED, requiring $\sim 10^{-20}$ J per gate.

C. The combined arithmetic warp bubble

A two-layer structure achieves both repulsion and decoupling:

- **Outer shell:** all primes π-rotated $\to G_{\text{eff}} = -2G$ (repulsive, pushes surrounding space).
- **Inner region:** 2 primes muted $\to G_{\text{eff}} = 0$ (ship floats, no tidal forces).
- **Transition wall:** metamaterial boundary between layers.

Total energy: $\sim 10^3$ J (wall structure), compared to $\sim 10^{47}$ J of exotic matter for standard Alcubierre.

D. Computational constraints inside the bubble

Muting $p = 2$ weakens binary arithmetic (parity-based error correction loses its algebraic foundation in $\mathbb{F}_2$). Inside a K-prime-muted bubble, computation must use a number base $b > p_K$ (the largest muted prime):

Primes muted	Largest p	Safe base
$\{2, 3\}$	3	5 (quinary)
$\{2, \ldots, 13\}$	13	17
$\{2, \ldots, 43\}$	43	47

Error correction over GF(47) or larger is standard in cryptographic applications (Reed-Solomon codes).

E. Entropy maintenance cost

The Landauer bound gives the minimum energy to maintain the muted state: $W \geq k_B T\,|\Delta S|$ per mode, where $\Delta S \approx 0.39$ (from BC entropy calculation). At 10 mK: $W \sim 10^{-8}\,\mathrm{J\,m^{-3}}$—negligible compared to the wall energy.

VIII. INFORMATION GEOMETRY OF THE $G = 0$ REGION

A critical question: does the speed of light change when $G_{\text{eff}} = 0$?

A. Light speed is unchanged (Connes distance proof)

For the Connes distance $d(x, y) = \sup\{|f(x) - f(y)| : \|[D, f]\| \leq 1\}$, a scalar perturbation $D \to D + V(x)$ gives $[D + V, f] = [D, f]$ (since $[V, f] = 0$ for multiplication operators). Therefore $d(x, y)$ is *unchanged*: $c_{\text{eff}} = c$ inside the $G = 0$ bubble.

This is confirmed by two independent methods: (1) the quantum speed limit (Mandelstam-Tamm) gives $v_{\text{muted}}/v_{\text{normal}} \sim O(1)$, and (2) the Fisher information metric is nearly invariant under muting.

B. Spectral dimension reduction on the bubble wall

The *spectral dimension* $d_S(t) = -2\,d(\log K)/d(\log t)$ reveals a striking asymmetry between the bubble layers.

For the Liouville vacuum (outer shell, $G = -2G$): the heat kernel $K_\pi(t) = \zeta(t)/\zeta(2t)$ gives $d_S \approx 0.4$ at UV scales, dropping *below 2*—a holographic dimension reduction.

Layer	d_S (intermediate t)	Causal structure
Interior ($G = 0$)	≈ 4	Standard 4D light cone
Outer shell ($G = -2G$)	≈ 2	2D holographic
Exterior (normal)	≈ 4	Standard 4D light cone

C. FTL mechanism via dimensional reduction

The ship is at rest inside the $G = 0$ region ($c_{\text{eff}} = c$, no FTL internally). The *bubble wall*, however, propagates on the $d_S \approx 2$ holographic surface of the Liouville shell, where the causal structure differs from the 4D light cone.

In 2D conformal field theory, the "speed limit" is set by the conformal structure, not by the 4D metric. The effective Virasoro central charge of the Liouville shell is $c_{\text{Vir}} = 12\,\zeta(2)/\zeta(4) \approx 18$, indicating a non-trivial 2D CFT.

The bubble velocity is determined by the 2D propagation dynamics, which is not constrained by the 4D light cone. This provides the arithmetic analog of the Alcubierre FTL mechanism: rather than bending 4D spacetime, the bubble wall undergoes *dimensional reduction* to a 2D surface with different causal constraints.

IX. THE DECISIVE EXPERIMENT

The regulator conjecture (Section V) makes a sharp, falsifiable prediction.

Setup. Three BAW crystals:

A. Uniform (DD): eigenvalues $\propto n^2$ for all n.

B. DN boundary: eigenvalues $\propto (2n - 1)^2$ (odd n only). Implements $p = 2$ muting.

C. DN + internal Bragg reflectors at $L/3$, $2L/3$: eigenvalues $\propto m^2$ for m not divisible by 2 or 3. Implements $p = 2, 3$ muting.

Measurement. Temperature-sweep differential Casimir energy extraction: $\Delta E = E(T_2) - E(T_1)$ for each crystal.

Prediction.

	Euler product	Mode counting
$\lvert E_C/E_A \rvert$	182	1/3

A factor of 546 separates the two predictions. This is easily distinguishable experimentally.

Internal Bragg reflectors. Mo/AlN Bragg stacks (5 pairs, 500 nm total) deposited at positions $L/3$ and $2L/3$ within a 142 μm quartz crystal. The acoustic impedance contrast (quartz: 15.2 MRayls, Mo: 63.1 MRayls) gives $> 99\%$ reflectivity per stack.

Cost: \$2,200 total (including custom crystal fabrication). **Timeline:** 4–6 months.

X. DISCUSSION

A. What the arithmetic structure provides

The four approaches share a common insight: the Euler product decomposition of $\zeta(s)$ makes the vacuum energy *multiplicatively decomposable* into independent prime channels. This is a structure that does not exist in standard QFT (where the vacuum is treated as a single entity).

If the regulator conjecture is confirmed experimentally, the implications extend beyond vacuum engineering:

- Dark energy ($\Omega_\Lambda = 2\pi/9$) and Casimir energy are different evaluations of the *same* L-function $\zeta_{\neg 2}$.
- The Euler product is not merely a mathematical identity but a physical decomposition of the vacuum into prime sectors.

- Each prime sector can be independently manipulated, with multiplicative effects on vacuum energy.

B. What remains unresolved

1. **Local vs. global muting:** Does local mode filtering change coupling constants? The BC framework suggests yes (L-function values change); standard QFT UV/IR separation suggests no. The experiment decides.

2. **Energy scaling:** The product $\delta g \times E_{\text{wall}} \sim \text{const}$ holds in both GR and the spectral approach. The BC phase transition offers some leverage but not unlimited. Practical warp remains energy-limited.

3. **Regulator conjecture:** Whether $\zeta_{\text{muted}}(-3)$ faithfully gives the physical Casimir energy for highly non-standard mode spectra is the central open question.

C. Regardless of warp

Even if the framework does not lead to practical warp drive, the experimental program has independent value:

- Verification of the arithmetic vacuum structure ($\zeta_{\neg 2}$ as physical vacuum).

- A new test of zeta-function regularization (the most widely used but least tested regularization in QFT).

- Connection between laboratory Casimir physics and cosmological dark energy via the same L-function.

- A novel propellantless thruster concept (β-pumping).

XI. CONCLUSION

We have shown that the Bost-Connes C*-dynamical system provides a unified framework connecting dark energy, Casimir physics, and vacuum engineering through the arithmetic of the Riemann zeta function.

Five distinct approaches to vacuum manipulation were developed: the spectral action approach provides conceptual clarity, β-pumping offers immediate experimental access, Euler product amplification gives the largest potential effect, inverse spectral design provides the cleanest implementation, and Dirac spectral modulation yields the most radical prediction—local control of the effective gravitational coupling, including $G_{\text{eff}} = 0$ (gravity nullification via 2-prime muting) and $G_{\text{eff}} = -2G$ (anti-gravity via all-prime π-rotation through the Liouville function).

The combined arithmetic warp bubble—an outer repulsive shell ($G = -2G$) surrounding a gravity-free interior ($G = 0$)—requires $\sim 10^3$ J of positive energy, 44 orders of magnitude less than Alcubierre's 10^{47} J of exotic matter.

All five approaches converge on a single decisive experiment: measuring the Casimir energy ratio in a 2-prime-sieved acoustic cavity. The predicted ratio (182:1 vs. 1:3) is large enough for unambiguous determination with modest equipment ($2,200, 4–6 months).

If confirmed, the result would establish the Euler product as a physical—not merely mathematical—decomposition of the quantum vacuum, opening a path from number theory to vacuum engineering.

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